

8TH Semester

CHEM-431

Analytical Sp-1

(2017, 2019, 2020, 2022)

SHORT QUESTIONS

1) What is the effect of temperature on conductance of solution?

The conductance of a solution generally increases with temperature. This effect is primarily attributed to the increase in the mobility of ions within the solution as temperature rises. In a solution, ions are surrounded by a shell of solvent molecules, and higher temperatures provide more thermal energy to these solvent molecules, leading to greater ion mobility. Consequently, the ions move more freely through the solution, resulting in increased electrical conductivity.

2) How sputtering takes place in glow discharge?

Sputtering in glow discharge, also known as cathodic sputtering occurs through a process of momentum transfer between energetic ions in the plasma and the atoms or molecules of a solid target material (the cathode). Sputtering in glow discharge involves the bombardment of a cathode material by energetic ions in plasma, resulting in the ejection of atoms or molecules from the cathode surface. This process is widely used in thin film deposition and surface modification applications in fields such as semiconductor manufacturing, materials science, and thin film technology.

3) Give the difference between Arc and Spark?

Arc	Spark
Arc discharge is a continuous, stable electrical discharge that forms a luminous bridge between two electrodes. It is characterized by	Spark discharge is a transient, intermittent electrical discharge characterized by a sudden and brief burst of energy. It occurs as a result of

sustained ionization and a relatively steady current flow.

Arc discharge can persist for extended periods, ranging from milliseconds to hours, depending on the conditions and the characteristics of the electrodes.

Arc discharge generates relatively high temperatures and releases significant amounts of energy due to the sustained current flow and ionization of the gas medium.

rapid breakdown and extinction of the gas medium between electrodes.

Spark discharge is short-lived, typically lasting only microseconds to milliseconds, as it quickly extinguishes after the breakdown event.

Spark discharge produces localized heating and energy release but at a much lower magnitude compared to arc discharge, as it involves a rapid discharge and quick extinction.

4) How oxygen interferes in polarography analysis? How to over-come it.

Oxygen interference in polarographic analysis can occur due to its reduction at the working electrode, leading to unwanted currents and distortions in the recorded polarographic curves. This interference is particularly common in the determination of analytes that undergo reduction at potentials close to that of oxygen reduction.

Method to Overcome Oxygen Interference: Oxygen scavengers, such as sodium sulfite (Na_2SO_3) or sodium dithionite ($\text{Na}_2\text{S}_2\text{O}_4$), can be added to the electrolyte solution to chemically remove dissolved oxygen. These compounds react with oxygen to form stable byproducts, preventing its reduction at the working electrode.

5) Distinguish between Specific and Molar conductance?

Specific Conductance

Specific conductance, also known as conductivity, is a measure of the ability of a solution to conduct electric current. It is expressed as the

Molar Conductance

Molar conductance is the specific conductance of a solution containing one mole of electrolyte per unit volume. It provides a measure of the

conductance of a solution normalized by the cell constant and the distance between the electrodes.

Units: Siemens per meter (S/m) or microsiemens per centimeter ($\mu\text{S}/\text{cm}$).

Formula: Specific conductance (κ) =
Conductance (G) / Cell Constant (K) *
Distance between electrodes (L)

conductivity per mole of the electrolyte and is affected by the concentration of the electrolyte.

Units: Siemens per meter per mole ($\text{S}\cdot\text{m}^2/\text{mol}$) or siemens per centimeter per mole ($\text{S}\cdot\text{cm}^2/\text{mol}$).

Formula: Molar Conductance
(Λ) = Specific Conductance (κ) / Molar
concentration (c).

6) What do you mean about residual current in polarography?

In polarography, residual current refers to the small steady-state current that persists after the diffusion-controlled reduction or oxidation of an analyte has reached equilibrium at the working electrode. This residual current is observed in the absence of any analyte signal and is primarily attributed to the continued reduction or oxidation of residual oxygen or other impurities in the electrolyte solution.

7) What are the advantages and disadvantages of conductometric titrations?

Advantages	Disadvantages
Wide Applicability: Conductometric titrations can be applied to a wide range of chemical reactions, including acid-base, redox, and precipitation reactions. This versatility makes them suitable for various analytical and research purposes.	Limited Selectivity: Conductometric titrations may lack selectivity in complex sample matrices containing multiple ions or interfering species.
High Sensitivity: Conductometric measurements are highly sensitive to changes in conductivity, allowing for the detection of subtle changes in analyte concentration during titration.	Electrode Polarization: Electrode polarization effects can occur during conductometric titrations, leading to deviations from ideal behavior and inaccuracies in conductivity measurements.
	Limited Range of Analytes: While conductometric titrations can be

This sensitivity enables precise endpoint determination and accurate quantification of analytes.

Rapid Analysis: Conductometric titrations often proceed rapidly, especially for reactions with significant conductivity changes near the equivalence point.

applied to various types of chemical reactions, they may not be suitable for all analytes. Certain analytes with low conductivity changes or those that do not produce ions in solution may not be amenable to conductometric titrations.

8) Give three characteristics of supporting electrolyte to be used in polarography?

The choice of supporting electrolyte in polarography is critical for achieving accurate and reliable results. Here are three important characteristics of a supporting electrolyte used in polarography:

Ionic Strength:

The supporting electrolyte should have sufficient ionic strength to enhance the conductivity of the solution without introducing excessive background current or interfering with the redox reactions of the analyte. Common supporting electrolytes used in polarography include potassium chloride (KCl), sodium sulfate (Na_2SO_4), and potassium nitrate (KNO_3), which provide suitable ionic strength for many applications.

Chemical Inertness:

The supporting electrolyte should be chemically inert towards the analyte and electrode materials to prevent unwanted side reactions that could interfere with the polarographic measurements.

Low Background Current:

A good supporting electrolyte should contribute minimal background current to the polarographic measurements to avoid obscuring the signal from the analyte.

Low background current reduces noise and interference, improving the sensitivity and precision of the polarographic technique.

9) Write basic principle of glow discharge tube?

Glow discharge tubes operate based on the principle of creating a low-pressure gas discharge between two electrodes. When a high voltage is applied across the electrodes in a low-pressure gas environment, it ionizes the gas, creating a plasma or glow discharge. This discharge generates excited atoms and ions, emitting characteristic spectral lines that can be analyzed to identify elements present in the gas. Glow discharge tubes are commonly used in spectroscopic techniques such as optical emission spectroscopy (OES) and mass spectrometry for elemental analysis.

10) What are limitations of Amperometry?

Amperometry has several limitations, including:

Limited Selectivity: Amperometry may lack selectivity in complex sample matrices, leading to interference from other electroactive species.

Electrode Fouling: Electrode fouling can occur due to the accumulation of reaction products or adsorbed species on the electrode surface, leading to drift in the measured current over time.

Baseline Drift: Changes in temperature, electrolyte composition, or electrode conditions can cause baseline drift in amperometric measurements, affecting the accuracy and precision of the results.

11) What is the effect of dilution of specific conductance?

Dilution of a solution decreases its specific conductance due to a decrease in the concentration of ions available for conduction. As the solution is diluted, the number of ions per unit volume decreases, resulting in reduced conductivity. This relationship is described by the equation: $\text{Specific Conductance} = \text{Conductivity} / \text{Concentration}$. Therefore, increasing dilution leads to a proportional decrease in specific conductance.

12) What is the basic principle of Anode Stripping Voltammetry?

Anode Stripping Voltammetry is a voltammetric technique used for the quantitative determination of metals or metal ions in solution. The basic principle involves the deposition of metal ions onto a working electrode (usually a mercury drop electrode) by applying a suitable negative potential. After deposition, the electrode potential is scanned in the positive direction, causing the desorption and oxidation of the deposited metal ions back to their original form. The resulting current peak is proportional to the concentration of the metal ion in the solution, allowing for its quantitative analysis.

13) What is ILKOVICH Equation? Give its significance.

The Ilkovich equation is used in polarography to relate the diffusion current to the concentration (c) of the analyte at the dropping mercury electrode (DME). The equation is:

$$I_d = k \sqrt{c}$$

Significance: The Ilkovich equation helps in determining the concentration of the analyte based on the diffusion current observed during polarographic measurements. It provides a quantitative relationship between the current and analyte concentration, aiding in the accurate determination of analyte concentrations in solution.

14) Give application of Conductometry in precipitation titrations?

- Conductometry is commonly used in precipitation titrations to monitor the formation of insoluble precipitates.
- As the titrant is added, it reacts with the analyte, leading to the formation of a precipitate.
- The conductivity of the solution decreases as the concentration of free ions decreases due to the formation of the precipitate.

- The endpoint of the titration is determined by monitoring the sharp decrease in conductivity, indicating the completion of the precipitation reaction.

15) What do you know by cyclic voltammetry?

Cyclic voltammetry is an electrochemical technique used to study the redox behavior of a solution or analyte.

In cyclic voltammetry, the potential of the working electrode is linearly swept back and forth between two limits at a constant scan rate. The resulting current response is recorded as a function of applied potential, generating a cyclic voltammogram. Cyclic voltammetry provides information about the redox potentials, kinetics, and reversibility of electrochemical reactions.

16) Give the changes in the conductance in the titration of mixture of HCL and Acetic Acid with NaOH?

- In the titration of a mixture of HCl and acetic acid with NaOH, the conductance initially decreases due to the neutralization of HCl with NaOH, forming a salt (NaCl).

After the equivalence point of HCl is reached, the conductance starts to increase due to the neutralization of acetic acid with NaOH, forming sodium acetate (NaCH₃COO). The conductance reaches a maximum at the equivalence point of acetic acid, indicating the completion of the titration.

17) What is Cell Constant? Give its relation between conductivity of cell and cell constant?

Cell constant (K) is a measure of the geometric characteristics of the conductivity cell and is defined as the ratio of the distance between the electrodes (L) to the area of the electrode (A).

$$\text{Cell Constant (K)} = L / A$$

The conductivity (κ) of the cell is directly proportional to the cell constant (K) and is inversely proportional to the resistance (R) of the cell. The relationship is given by:

$$\kappa = 1/R \cdot K$$

18) What different arc and spark sources used in Spectroscopy?

In spectroscopy, arc and spark sources are used to produce high-temperature plasma for excitation and atomization of analyte atoms. Arc sources use an electric arc discharge between two electrodes to generate the plasma, while spark sources use a series of rapid electrical discharges (sparks) between electrodes. Arc sources are suitable for solid samples and provide high temperatures, while spark sources are more versatile and can analyze solid, liquid, and gaseous samples.

19) Give three advantages of DME (Dropping Mercury Electrode)?

High Sensitivity: DMEs provide high sensitivity due to the small drop size and the high surface-to-volume ratio.

Wide Potential Range: DMEs can be operated over a wide potential range, allowing for the investigation of various redox reactions.

Rapid Response: DMEs offer rapid response times, making them suitable for kinetic studies and fast scan voltammetry techniques.

20) What are three basic mechanisms of mass transport in voltammetry?

Migration: Movement of ions towards the electrode under the influence of an electric field.

Diffusion: Movement of ions or molecules through the solution due to concentration gradients.

Convection: Movement of solution due to bulk flow or stirring, enhancing mass transport to the electrode surface.

21) Differ between cathodic current and anodic current?

Cathodic Current	Anodic Current
• Represents the flow of electrical current towards the working electrode during reduction reactions. • In polarography, cathodic current corresponds to the reduction of analyte species at the working electrode.	• Represents the flow of electrical current away from the working electrode during oxidation reactions. • In polarography, anodic current corresponds to the oxidation of analyte species at the working electrode.
• Cathodic current is typically observed as a negative peak in the voltammogram.	• Anodic current is typically observed as a positive peak in the voltammogram.

22) Describe role of inert supporting electrolyte in polarography?

An inert supporting electrolyte is used in polarography to provide conductivity and maintain a stable ionic environment without interfering with the redox reactions of the analyte.

- The supporting electrolyte helps to establish a suitable ionic strength in the solution, facilitating ion transport to the electrode surface and ensuring a stable double layer.
- By maintaining a constant conductivity and ionic strength, the inert supporting electrolyte helps to improve the reproducibility, sensitivity, and accuracy of polarographic measurements.

Additionally, inert supporting electrolytes minimize the risk of unwanted side reactions or interference from the electrolyte itself, allowing for reliable determination of analyte concentrations.

23) Describe disadvantages of Automation Process?

Initial Cost: Automation systems often require significant initial investment in equipment, software, and infrastructure, which may be cost-prohibitive for some laboratories or industries.

Complexity: Implementing automation processes may require extensive planning, programming, and optimization, which can be complex and time-consuming.

Maintenance Requirements: Automated systems require regular maintenance, calibration, and troubleshooting to ensure proper functioning, which may increase operational costs and downtime.

Training Needs: Operating and maintaining automated systems may require specialized skills and training, necessitating investment in workforce development and education.

24) What is the principle of Amperometry?

Amperometry is an electrochemical technique based on the measurement of current resulting from an electrochemical reaction occurring at an electrode. The principle involves applying a constant potential (voltage) between the working electrode and a reference electrode in a solution containing the analyte of interest. As the analyte undergoes oxidation or reduction at the working electrode, a current proportional to the rate of the electrochemical reaction is generated and measured. By monitoring the current over time, the concentration of the analyte can be determined quantitatively.

25) How glow discharge atomization takes place?

In glow discharge atomization, a low-pressure gas discharge is established between two electrodes in a controlled environment. When a high voltage is applied across the electrodes, it ionizes the gas, creating plasma or glow discharge. The high temperature of the plasma excites and atomizes the sample atoms or molecules present in the gas phase, forming free neutral atoms or

ions. These neutral atoms or ions can then be analyzed spectroscopically to determine the elemental composition of the sample.

26) Differ between Fixed and Flexible Automation?

Fixed Automation	Flexible Automation
• In fixed automation, the equipment or system is designed to perform a specific set of tasks or operations in a predetermined sequence. • The configuration and capabilities of fixed automation systems are static and cannot be easily modified or adapted to different tasks or production requirements. • Fixed automation systems are often used for high-volume production of standardized products with consistent processes and minimal variation. • Examples include assembly lines, conveyor systems, and dedicated manufacturing machinery.	• In flexible automation, the equipment or system is designed to handle a variety of tasks or operations with the ability to adapt to changing requirements or conditions. • Flexible automation systems are characterized by their versatility, modularity, and ability to be reconfigured or reprogrammed for different tasks or product variations. • These systems often utilize advanced control systems, robotics, and programmable logic controllers (PLCs) to enable rapid changeovers and customization. • Flexible automation is commonly used in industries with diverse product lines, custom manufacturing, or low to medium-volume production.

27) How conductance is measured?

Conductance is measured using a conductometer by applying a known voltage across electrodes immersed in a sample solution and measuring the resulting current flow. The conductivity of the solution is calculated based on the

conductance and cell constant, providing valuable information about the sample's electrical properties.

The conductometer measures the electrical conductance of the sample solution by applying a voltage across the electrodes and measuring the resulting current flow. The conductivity (κ) of the solution is calculated using Ohm's law:

$$\kappa = G/L$$

28) Write down the principle of conductometric titrations?

- Conductometric titrations are based on the principle of measuring changes in the conductivity of a solution as a titrant is added to a sample solution containing an analyte.
- The titrant reacts with the analyte to form a product, resulting in a change in the concentration of ions in the solution and, consequently, a change in conductivity.
- The endpoint of the titration is determined by monitoring the conductivity of the solution. A sharp change in conductivity indicates that the reaction between the titrant and analyte is complete.
- Conductometric titrations are commonly used for acid-base titrations and precipitation titrations, where the formation or disappearance of ions during the reaction leads to changes in conductivity.

29) Give disadvantage of anodic stripping voltammetry?

One disadvantage of anodic stripping voltammetry is the susceptibility to electrode fouling. During the deposition step, metal ions are plated onto the working electrode surface. However, other species present in the solution may also adsorb onto the electrode, leading to fouling. This fouling can interfere with subsequent measurements by altering the electrochemical response of the electrode or masking the signals of interest.

30) Differ between migration current and residual current?

<u>Migration Current:</u>	<u>Residual Current:</u>
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- | | |
|---|---|
| <ul style="list-style-type: none"> • Migration current refers to the current resulting from the movement of ions towards the electrode under the influence of an electric field during voltammetry. • It is proportional to the concentration of the analyte and is observed as a current peak in the voltammogram. | <ul style="list-style-type: none"> • Residual current refers to the steady-state current that persists after the completion of an electrochemical reaction, typically in polarography or voltammetry. • It is often attributed to side reactions, electrode polarization, or residual analyte species in the solution and is observed as a baseline offset or steady current level in the absence of analyte. |
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31) Describe amperometric titrations involving electro-reducible vs electro-oxidizable?

In amperometric titrations involving electro-reducible species, the analyte is reduced at the working electrode, leading to a measurable current response.

Conversely, in amperometric titrations involving electro-oxidizable species, the analyte is oxidized at the working electrode, resulting in a detectable current change. The titrant is added incrementally to the analyte solution, and the current response is recorded as a function of titrant volume. The endpoint of the titration is determined based on a significant change in the current, indicating the completion of the reaction between the titrant and analyte.

32) What is the effect of charge and ion on conductivity?

Charge: The conductivity of a solution is influenced by the concentration and mobility of charged ions present in the solution. Higher concentrations of charged ions generally lead to higher conductivity due to increased charge carriers.

Ion: The conductivity of a solution also depends on the nature of the ions present. Strong electrolytes that dissociate completely into ions in solution

exhibit higher conductivity compared to weak electrolytes with partial ionization.

33) What is meant by arc and spark ablation?

- **Arc Ablation:** Arc ablation involves the use of a high-voltage electric arc discharge to vaporize and atomize a solid sample, typically a metal, for analysis by spectroscopic techniques such as atomic emission spectroscopy (AES) or inductively coupled plasma mass spectrometry (ICP-MS).
- **Spark Ablation:** Spark ablation utilizes a series of rapid electrical discharges (sparks) between two electrodes to vaporize and atomize the sample material. This technique is commonly used in spark emission spectroscopy (Spark-OES) for the elemental analysis of solid samples.

34) What are advantages of Amperometric Titrations?

High Sensitivity: Amperometric titrations offer high sensitivity, allowing for the detection of analytes at low concentrations.

Wide Applicability: Amperometric titrations can be applied to a wide range of analytes, including both electroactive and non-electroactive species.

Rapid Response: Amperometric measurements provide rapid results, enabling real-time monitoring of reactions and efficient titration processes.

Automation Compatibility: Amperometric titrations can be easily automated, reducing manual labor and increasing throughput in analytical laboratories.

35) What do you know about polarographic maxima?

Polarographic maxima refer to the peaks or maxima observed in the polarogram obtained during polarographic analysis. These maxima correspond to characteristic changes in the current response observed as a function of the applied potential. Polarographic maxima are often used to identify and characterize electrochemical processes occurring at the working electrode, such as reduction or oxidation reactions.

36) Describe briefly two steps in stripping voltammetry?

Deposition Step: In the deposition step, the analyte of interest is deposited onto the surface of the working electrode through an electrolytic process. This typically involves applying a constant potential to the working electrode to reduce the analyte to its metallic form or to preconcentrate it onto the electrode surface.

Stripping Step: After deposition, the potential applied to the working electrode is scanned, causing the oxidized form of the analyte to be stripped from the electrode surface and detected as a current peak. The height of the current peak is proportional to the concentration of the analyte in the sample.

37) Describe two factors that contribute to the chemical reactions in Polarography?

Electrode Kinetics: The rate of electrode reactions, including the reduction or oxidation of analyte species, depends on factors such as electrode material, surface area, and potential applied.

Solution Chemistry: Solution properties, including pH, temperature, and presence of complexing agents or supporting electrolytes, can influence the nature and kinetics of chemical reactions occurring at the electrode interface.

38) How the conductance of the solution changes under influence of Temperature and Concentration of ions?

Temperature: Conductance generally increases with increasing temperature due to the increased mobility of ions in solution. Higher temperatures enhance the thermal energy of ions, allowing them to move more freely and thereby increasing conductivity.

Concentration of Ions: Conductance is directly proportional to the concentration of ions in solution. Higher concentrations of ions result in higher conductance due to increased charge carriers available for conduction.

39) What are advantages of conductometric titration over the ordinary titrations?

Direct Detection: Conductometric titrations allow for direct detection of the endpoint based on changes in conductivity, eliminating the need for visual indicators or complex instrumentation.

Wide Applicability: Conductometric titrations can be applied to a wide range of acid-base, redox, and precipitation reactions, making them versatile for various analytical applications.

High Sensitivity: Conductometric titrations offer high sensitivity, enabling the detection of subtle changes in analyte concentration.

THE END

Cyanide